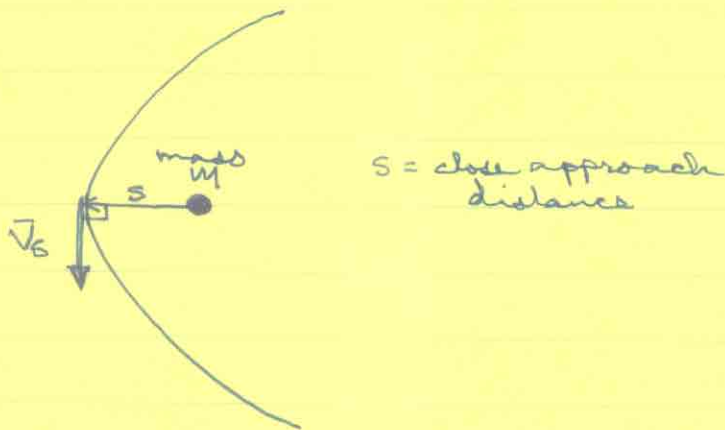


08/23/07

①

Hills' Mechanism

Kick to a parabolic orbit:



• let $E(\text{initial}) = 0 = \frac{1}{2} v_s^2 - \frac{GM}{s} \Rightarrow v_s = \left(\frac{2GM}{s} \right)^{1/2}$ [escape]

• give the particle a kick velocity $\vec{u}$ at pericenter

$$\Rightarrow E(\text{new}) = \frac{1}{2} (\vec{v}_s + \vec{u})^2 - \frac{GM}{s} = \frac{1}{2} v_s^2 - \frac{GM}{s} + \vec{v}_s \cdot \vec{u} + \frac{1}{2} u^2$$

suppose that $\vec{v}_s \cdot \vec{u} = v_s u$ (parallel kicks) and

$$u/v_s \ll 1$$

$$\Rightarrow E(\text{new}) \approx v_s u$$

$$= \frac{1}{2} v_\infty^2 \Rightarrow v_\infty^2 = 2 v_s u$$

$$\Rightarrow v_\infty = \sqrt{\frac{2u}{v_s}} \cdot v_s$$

Now let's put this to use ...

Let $M \approx 10^6 M_\odot$ (supermassive BH)

$$\Rightarrow v_s \approx 4200 \left(\frac{M_b}{52} \right)^{1/2} \text{ km s}^{-1}$$

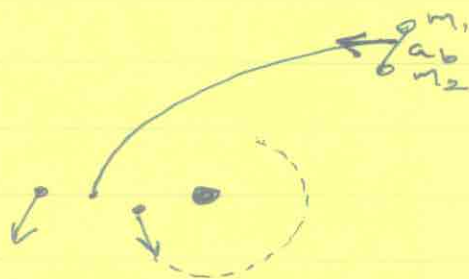
$\xrightarrow{10^6 M_\odot}$
 $\xrightarrow{100 \text{ AU}}$

$$\text{If } u = 100 u_2 \text{ km s}^{-1}$$

$$\Rightarrow v_\infty \approx [2 \cdot 100 \cdot 4200 u_2 \left(\frac{M_b}{52} \right)^{1/2}]^{1/2}$$

$$\approx 10^3 u_2^{1/2} \left(\frac{M_b}{52} \right)^{1/4} \text{ km s}^{-1}$$

Astrophysical context :



one star captured, one ejected ;
if, $s \lesssim a \left(\frac{M}{m_b} \right)^{1/3}$

$$m_b = m_1 + m_2$$

tidal impulse gives binary components a relative kick speed of $\sim v_b$ near pericenter of orbit around BH

$$\Rightarrow m_1 \text{ gets kick } \sim \frac{m_2}{m_b} v_b \sim \underline{u}.$$

(3)

$$\Rightarrow s \approx a \left(\frac{M}{m_b} \right)^{1/3}$$

$$\Rightarrow \frac{M}{s} \approx \frac{m_b}{a} \left(\frac{M}{m_b} \right)^{2/3}$$

$$\Rightarrow v_s \approx \sqrt{2} \cdot v_b \left(\frac{M}{m_b} \right)^{1/3} ; v_b \approx \left(\frac{Gm_b}{a} \right)^{1/2}$$

$$u_1 \approx \frac{m_2}{m_b} v_b ; v_s \approx v_b \sqrt{2} \left(\frac{M}{m_b} \right)^{1/3}$$

$$\Rightarrow \sqrt{2u_1 v_s} \approx \left[2 \frac{m_2}{m_b} v_b \cdot v_b \cdot \sqrt{2} \left(\frac{M}{m_b} \right)^{1/3} \right]^{1/2}$$

$$\approx \underbrace{1.7}_{\sim 1} \underbrace{\left(\frac{M_2}{m_b} \right)^{1/2} \left(\frac{M}{m_b} \right)^{1/6}}_{\sim 10 \frac{M}{m_b}^{1/6}} \cdot v_b$$

a tight binary may have $a \sim 1 R_\odot \Rightarrow v_b \sim 400-500 \text{ km s}^{-1}$

$\Rightarrow \sim 4000-5000 \text{ km s}^{-1}$ for v_∞ .

But, note $s \approx 1 R_\odot \left(\frac{10^6}{1} \right)^{1/3} \sim 100 R_\odot \sim \frac{1}{8} \text{ AU}$

$$r_{\text{Sch}} \sim 3 \cdot 10^6 \text{ km } M_\odot \sim 0.04 \text{ AU}$$

$\Rightarrow \frac{2M}{s} \sim \frac{1}{10} \Rightarrow \text{GR effects becoming important!}$